REVIEW ARTICLE

Vitamin D deficiency, skin, and sunshine: A review

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Abstract

The sunshine vitamin - "Vitamin D" is a subject of interest that attracted researcher's attention in the past few decades. Vitamin D is a fat-soluble vitamin and also a nutrient which increases the calcium absorption and plays an important part in the maintenance of body's immune system and bone formation. However, the deficiency of Vitamin D, also known as hypovitaminosis D, will affect various body parts such as the brain, heart, muscle, immune system, and bones. Thus, the deficiency leads to severe conditions such as osteopenia, osteoporosis, rickets (in children), hypertension, fractures and falls in adults, cancers, diabetes, autoimmune diseases, infections, and neurological disorders. An effective approach for the deficiency prevention is the thorough understanding of Vitamin D sources, Vitamin D serum levels, and deficiency symptoms, linked with different pathological conditions, requirement and maintenance in the body, sunlight exposure duration, and effective treatment dose. Therefore, the present review will lay an emphasis on the role of Vitamin D, the reasons for its deficiency, available sources, and treatment options reported so far.

Key words: Skin, sunlight, Vitamin D, Vitamin D2, Vitamin D3

INTRODUCTION

Nowadays, Vitamin D deficiency affects 50% of the population worldwide. The main reason for its deficiency is the change of lifestyle, i.e. increase in sunscreen use and environmental factors such as reducing the time for exposure to sunlight, which is required for ultraviolet (UV) B rays to induce the production of Vitamin D in the skin. The people who are having black skin require more exposure to the skin for the production of Vitamin D as compared to the white ones.[1] Insufficient Vitamin D will lead to the condition of hypovitaminosis D and the latter is linked with numerous conditions of the body; these are cancer, heart diseases, Type 2 diabetes, probability of stress fractures, autoimmune disease, influenza, and depression.[2]

Vitamin D - Synthesis and Metabolism

Vitamin D is available in two forms: Vitamin D_3 (cholecalciferol) and Vitamin D_2 (ergocalciferol). Vitamin D_3 is a natural occurring form and produced in skin cells from 7-dehydrocholesterol underneath the UV light, and Vitamin D_2 is produced from the natural sterol, i.e. ergosterol.

Vitamin D_3 is metabolized in a two-step non-catalytic process favored by UV light and level of skin pigmentation, into 25-hydroxyvitamin D (25OHD) and then to 1,25-dihydroxyvitamin D (1,25$(OH)_2$D) which is a hormonal form and increases the intestinal calcium absorption, while Vitamin D_2 is metabolized to 25(OH)D in the liver and then to 1,25$(OH)_2$D in the kidneys from ergosterol,[3,4] as shown in Figure 1. The presence of melanin in the skin cells limits the Vitamin D_3 production by blocking the UV rays which affect in the synthesis process. Hence, wearing full sleeves clothes and use of sunscreen too confine its production. In our body, some tissues such as breast, macrophages, colon,

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Received: 04-06-2018
Revised: 18-06-2018
Accepted: 25-06-2018